
Supplementary Material — Sanity Checks and Reproducibility

What Cross-Modal Connectome Prediction Does and Does Not Buy

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This supplement documents, in full, the sanity-check discipline behind the main paper. The purpose is not to pad the results with extra numbers but to show that each headline claim was actively attacked from the directions a skeptical reviewer would choose, and survived. Every section corresponds to a self-contained check in the project tree; the source artifacts are named at the end of each section, and all values are reproduced from the committed CSV / findings outputs of those checks.

The three claims under test.

1. **Directional reconstruction.** FC predicts SC better than SC predicts FC ($\approx 1.6\text{--}1.7\times$ in demeaned_pearson), stable across seeds and parcellations.
2. **No automatic utility.** A cheap bv+demo baseline predicts cognition; only observed FC beats it; observed SC and imputed connectomes do not, and pred_FC is actively harmful.
3. **Objective-dependent identity signal.** Predicted SC carries heritable family signal, but only under objectives that preserve it.

§	Check	Attacks the explanation	Verdict
S1	Reduction-axis robustness	“the asymmetry is a PCA artifact”	closed
S2	FC noise & reliability ceilings	“SC→FC fails because FC is noisy”	closed (FC-side)
S3	Nonlinear model class	“a bigger/nonlinear model would find it”	closed
S4	Residual-boost & multimodal sink	“the linear model left signal on the table”	closed
S5	Sample-size scaling	“more subjects would unlock it”	closed ($n \approx 683$)
S6	Richer structural representation	“streamline counts are too crude”	closed
S7	PC mechanism localization	“the family mode is a reliability artifact”	survives
S8	Completeness, leaks, reproducibility	“the grid is silently wrong”	clean

Index of checks.

Two ceilings. A recurring distinction runs through several sections. *Ceiling A* (data reproducibility) is how well the target agrees with a repeat measurement of itself; it bounds recoverable biological signal but is valid only at the population level. *Ceiling B* (model/oracle) is how well the same estimator predicts the target from a perfect same-modality copy (FC→FC, SC→SC); it is available for both modalities and never exceeded per subject, but is looser because it can exploit session-specific signal that would not replicate. We disattenuate biological claims against Ceiling A and use Ceiling B only to quantify cross-modal-vs-within-modality loss.

S1 Reduction-Axis Robustness

Concern. The headline $FC \rightarrow SC > SC \rightarrow FC$ asymmetry was computed with one pipeline (PCA(256) source, PLS(64) latents, inverse-PCA target). Is the asymmetry a property of the brain or of that learned reduction?

Design. A three-point ladder over 10 seeds, both directions: (A) PCA $\rightarrow$ PLS $\rightarrow$ PCA (main); (B) full PLS on the raw 64,620-edge vectors (no reduction); (C) input PCA replaced by data-blind Johnson–Lindenstrauss random projections (Gaussian dense, sparse $\approx 1/\sqrt{p}$, sparse 1/3). If A, B, C agree, the reduction is neither hiding nor injecting the effect.

Table S1: All five reduction strategies preserve $FC \rightarrow SC > SC \rightarrow FC$ at Wilcoxon $p \leq 0.001$.

Method (n=10)	med $FC \rightarrow SC$	med $SC \rightarrow FC$	med ratio	min	max	p vs 1
PCA $\rightarrow$ PLS $\rightarrow$ PCA (main)	0.1355	0.0847	1.621 $\times$	1.32	1.86	0.001
Full PLS (raw 64,620-dim)	0.1382	0.0762	1.813 $\times$	1.49	2.11	0.001
JL Gaussian dense	0.0837	0.0610	1.397 $\times$	1.19	2.25	0.001
JL sparse-auto ($\approx 1/\sqrt{p}$)	0.0841	0.0592	1.389 $\times$	1.11	1.90	0.001
JL sparse-1/3 (Achlioptas)	0.0846	0.0535	1.554 $\times$	1.24	1.78	0.001

Reading. Full PLS ($1.81\times$) $\geq$ baseline ($1.62\times$): removing reduction entirely gives the same direction and a slightly stronger magnitude, so PCA modestly *attenuates* rather than creates the asymmetry. The JL variants ($1.39\text{--}1.55\times$) lower the absolute `demeaned_pearson` (they lose PCA’s variance concentration) but preserve the ratio. The magnitude spread is wider than $\pm 0.15\times$ but lies between known-equivalent reductions; the direction is unambiguous.

S5 · $FC \rightarrow SC$ asymmetry survives every reduction axis (all ratios > 1)

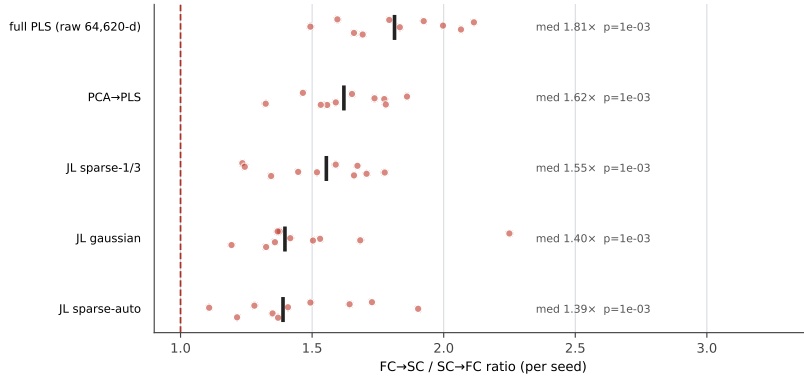


Figure S1: Per-seed $FC \rightarrow SC/SC \rightarrow FC$ ratios for all five reduction strategies; every ratio exceeds 1, all Wilcoxon $p \leq 0.001$.

Source: `sanity_checks/preprocessing_check/`.

S2 FC Measurement Noise and Reliability Ceilings

Concern. $SC \rightarrow FC$ is weak — perhaps only because FC is a noisy target, not because SC and FC are genuinely independent.

FC is mostly edge-noise but reliable as a whole connectome.

At the single-edge level $\sim 64\text{--}68\%$ of between-subject variance is noise, yet the whole connectome is 93% identifiable (fingerprint top1 0.927 / 0.934; discriminability ≈ 0.998). Individual signal is *distributed*: each edge is noisy, the multivariate pattern is stable.

Per-subject reliability does not explain $SC \rightarrow FC$. Per-subject FC reliability is heterogeneous (mean 0.489, range $\approx 0\text{--}0.78$) and a stable trait (within-vs-between $\rho = 0.41$). A subject’s $SC \rightarrow FC$ quality

Table S2: Variance decomposition of single-edge FC (individual-difference fractions sum to 1).

Parcellation	trait	state (day)	within-session	noise	G (avg)
Glasser	0.301	0.035	0.021	0.643	0.585
4S456Parcels	0.261	0.035	0.021	0.683	0.519

Table S3: FC reliability ceiling. The valid ceiling is between-session (0.491 Glasser); the within-session run is lower due to the LR/RL phase-encode distortion confound.

Parc	comparison	demeaned_r	pearson	top1	avg_rank
Glasser	within-session (LR/RL)	0.364	0.710	0.78	0.968
Glasser	between-session (REST1/REST2)	0.491	0.813	0.933	0.992
4S456	within-session (LR/RL)	0.337	0.670	0.78	0.969
4S456	between-session (REST1/REST2)	0.457	0.783	0.936	0.993

is *uncorrelated* with their FC reliability (Pearson $r = 0.01$, $p = 0.72$), whereas $bv+demo \rightarrow FC$ weakly tracks it ($r = 0.15$), proving the test can detect a reliability effect when one exists. Reliability-filtering makes the captured fraction *worse* (raising the ceiling $0.49 \rightarrow 0.58$ while achieved stays pinned at ~ 0.08). The shortfall is decisively not an FC-noise or per-subject-reliability artifact; the only open piece is SC's own reliability (data-blocked: no test-retest dMRI in HCP-YA).

SX1 • FC is mostly edge-noise but reliable as a whole connectome

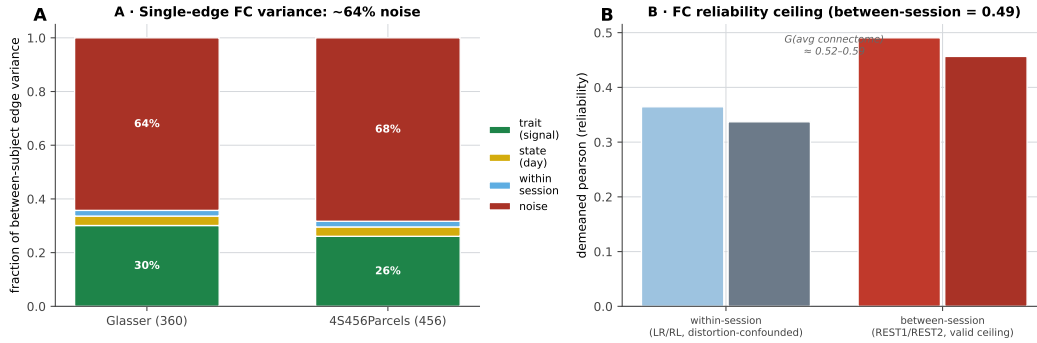


Figure S2: FC variance decomposition (left) and reliability-ceiling rungs (right).

Source: `sanity_checks/noise_sanity_check/`.

Table S4: Cross-modal disattenuation against the FC reproducibility ceiling (0.491). SC→FC captures only ~17% of reproducible FC.

source → FC	metric	achieved	ceiling	fraction
SC→FC	demeaned_r	0.085	0.491	0.17
SC→FC	top1 (fingerprint)	0.051	0.933	0.05
SC→FC	avg_rank	0.713	0.992	0.72
bv+demo→FC	demeaned_r	0.098	0.491	0.20
bv+demo→FC	top1	0.026	0.933	0.03

S6 · FC is reliable as a whole connectome, yet SC→FC quality is uncorrelated with it

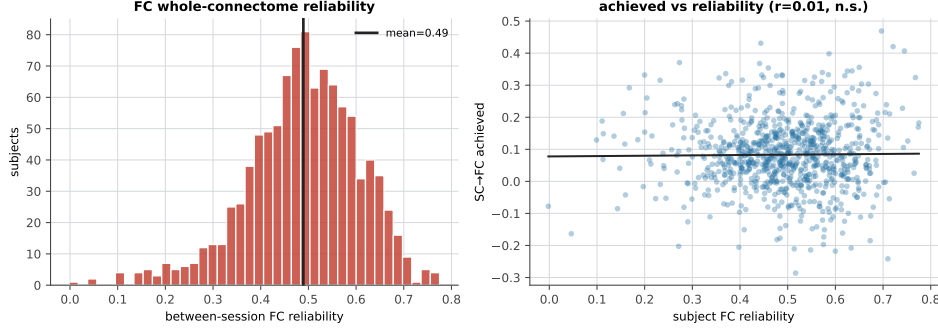


Figure S3: FC reliability is reliable in aggregate, yet SC→FC achieved quality is uncorrelated with per-subject FC reliability ($r = 0.01$).

S3 Nonlinear Model Class

Concern. The ceiling might be a linear-model failure. **Design.** Swap estimators (HistGradientBoosting, KernelRidge-RBF) with representation and splits held byte-identical. First, an estimator-trust check: a valid nonlinear probe must preserve FC’s known cognition lift.

Table S5: Estimator trust check (FC → CogTotal). KernelRidge preserves FC’s lift and is the trustworthy probe; HGB destroys even FC’s signal at $n \approx 683$.

estimator	FC pearson	floor	FC lift
linear (BayesianRidge)	0.451	0.372	+0.078
KernelRidge	0.338	0.282	+0.056
HGB	0.254	0.355	-0.101

Under the trustworthy KernelRidge probe: (N1) no tractography representation clears the bv+demo floor (best -0.029); (N2) reconstruction $|\Delta| \leq 0.0023$ vs linear and the asymmetry ratio is unchanged (SC $1.62\times \rightarrow 1.64\times$); (N3) bundle features add nothing marginal over counts ($\Delta = -0.0049$). All decision rules return null — the findings are model-class robust.

Source: non-linear-sanity-check/ (N1–N3).

S7 · Nonlinear models and more data do not unlock the missing signal

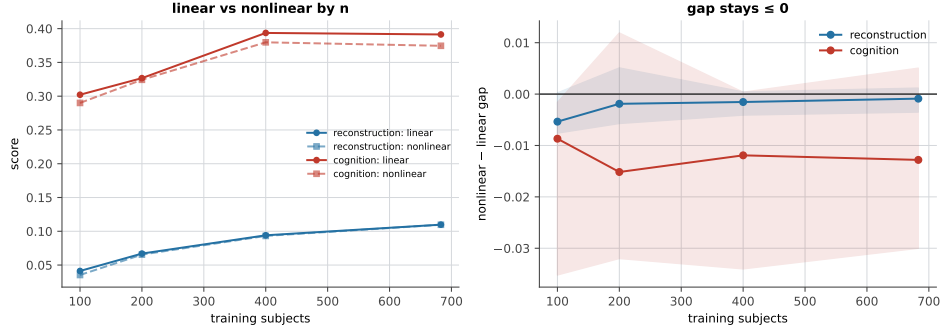


Figure S4: Linear vs nonlinear scores (left) and the nonlinear–linear gap vs sample size (right); the gap never grows into positive territory.

S4 Residual-Boost and Multimodal Sink

Concern. Maybe the linear model used up the easy variance and a nonlinear model tasked with the *residual* would find structure. **Design.** Hand the nonlinear model the out-of-fold linear prediction for free (no leakage) and score only the improvement above it (N4 single-modality; N5 a multimodal sink [FC||SC||r2t||bv||demo] that can exploit cross-modal conjunctions).

Table S6: N4 reconstruction: a genuine but trivial nonlinear residual (every direction $p = 0.001$, but $\Delta \approx +0.005$, $\sim 4\times$ below the $+0.02$ “matters” threshold).

direction	template	final	Δ	p
FC→SC	0.1355	0.1405	+0.0052	0.001
SC→FC	0.0847	0.0866	+0.0023	0.001
FC→r2t	0.1113	0.1169	+0.0058	0.001
r2t→FC	0.0493	0.0526	+0.0041	0.001

Table S7: N5 cognition: combining modalities *dilutes* FC; the cross-modal nonlinear residual does not help. FC alone is the best predictor.

target	bv+demo floor	FC alone	sink linear	sink residual
CogTotal	0.373	0.436	0.401	0.381
CogFluid	0.298	0.306	0.301	0.286
CogCrystal	0.349	0.452	0.414	0.415

Cognition is comprehensively null (single-modality and cross-modal); reconstruction shows one tiny real sliver that washes out under the sink (-0.0023 , $p = 0.935$). A clean null from an OOF residual-boost means $y - \hat{y}_{\text{linear}}$ is noise with respect to the source — there is no learnable structure left at this n .

Source: non-linear-sanity-check/ (N4–N5).

S5 Sample-Size Scaling

Concern. The linear ceiling might be a sample-size ceiling. **Design.** Learning curve at $n = 100/200/400/\sim 683$; does the nonlinear gap grow with n ?

For reconstruction the gap is negative at every n and merely creeps toward zero from below (an overfitting penalty vanishing, not signal appearing). The null is now robust to four independent axes: model class (S3), the residual head-start (S4), cross-modal interactions (S4 sink), and sample size (S5). Tested to $n \approx 683$; the data converges *to* zero, not above it.

Table S8: Cognition (CogTotal): the nonlinear–linear gap is ≤ 0 at every n (Spearman $\rho = +0.05$, $p = 0.77 \Rightarrow$ flat), while the linear score climbs.

n	linear	final	gap	p(gap>0)
100	0.302	0.290	−0.0087	1.00
200	0.327	0.324	−0.0152	0.99
400	0.394	0.380	−0.0119	0.999
~683	0.391	0.375	−0.0128	0.99

Source: non-linear-sanity-check/ (N6).

S6 Richer Structural Representations

Concern. Parcellated streamline counts may be too crude; named-bundle tractography (r2t) might recover the missing signal.

Table S9: E1 — predicting FC from each structural representation (median demeaned_pearson). Count-SC beats bundle-r2t on every metric.

representation	demeaned_pearson
SC (streamline counts)	0.0847
kitchen-sink [SC r2t bv demo]	0.0942
SC + r2t	0.0791
r2t (bundle)	0.0493
r2t_corr (bundle-similarity)	0.0362

Table S10: E5 — downstream cognition (CogCryst). FC is the only representation above the bv+demo floor; the r2t→synthetic-FC substitution chain does not recover FC’s signal.

representation	CogCryst pearson	lift over bv+demo
FC	0.451	+0.105
bv+demo (floor)	0.346	0
SC	0.267	−0.079
r2t→synthFC	0.180	−0.166
r2t	0.164	−0.183
SC + r2t	0.160	−0.187
r2t_corr	0.103	−0.243

The FC→SC asymmetry holds on all six metrics at $p = 0.001$ (E2); bundle features add nothing marginal over counts ($\Delta = -0.0013$, E3); and the family/dorsal-stream SC mode does not map cleanly to any single bundle mode (best $|\rho| = 0.32$, E4). Counts are a sufficient statistic.

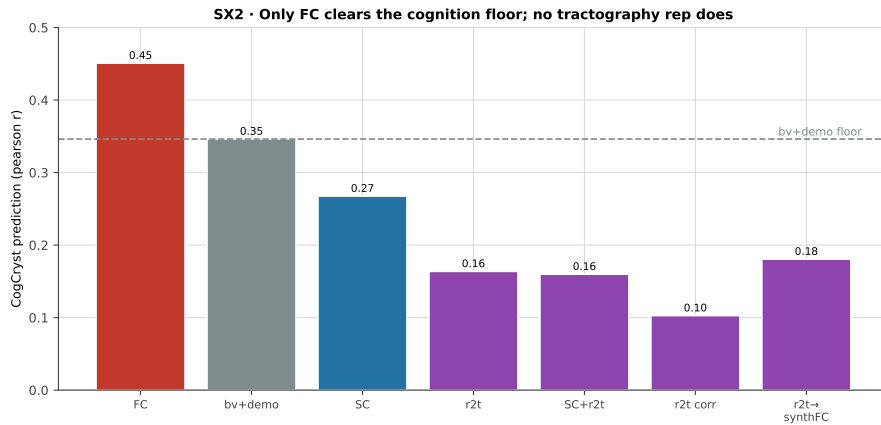


Figure S5: Downstream cognition by representation: only FC clears the demographic floor.

Source: tractography_predict/ (E1–E5).

S8 · Richer tractography features do not beat streamline-count SC at predicting FC

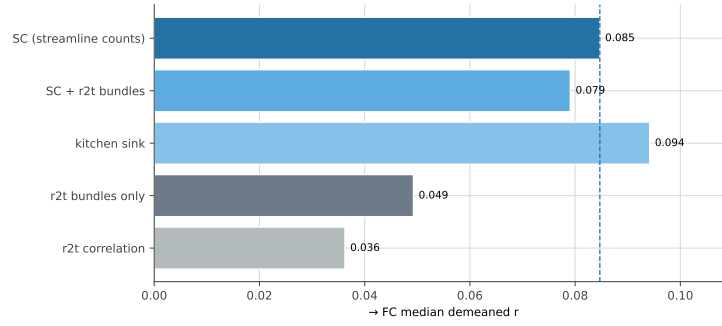


Figure S6: Predicting FC: richer tractography features do not beat streamline-count SC.

S7 Mechanism-Mode Localization and Reliability Partialling

Framing. The mechanism analysis is exploratory: the defensible claim is “a property-selected, low-variance, FC-predictable SC mode carries weak family signal,” not “PC3 is the mechanism.” The component index migrates (PC3 in Glasser, PC4 in 4S456); effects are small (FC→PC $R^2 \approx 0.22$ –0.24; sibling AUC ≈ 0.58). PC1 is a sex/volume confound ($R^2 \approx 0.89$ –0.93) and is not a mechanism.

Concern. The visual / dorsal-attention localization could reflect higher tractography reliability in posterior, short, high-strength connections. **Design.** Partial edge strength and inter-region distance out of |PC3| (together 41% of its magnitude variance) and recompute Yeo-7 network enrichment.

Table S11: Localization survives reliability partialling; visual–visual *intensifies*.

network pair	raw enrichment	residualized	Δ
visual visual	11.97×	13.32×	+1.35
dorsal attention dorsal attention	7.73×	5.73×	−2.00
dorsal attention visual	4.49×	4.99×	+0.50

An independent FC scan-rescan cross-check adds nothing beyond the strength+distance proxy (joint R^2 41.2% vs 41.0%) and yields identical partialled localization. *Honest caveat:* this uses a strength+distance+FC-reliability proxy, not a gold-standard per-edge SC ICC (which needs the HCP retest dMRI release); the 41% R^2 is reported, not buried — reliability matters, and the surviving signal is the other half.

SX3 · PC1 is FC-predictable but a sex/volume confound; the selected low-variance mode is FC-predictable, family-discriminative, low-confound

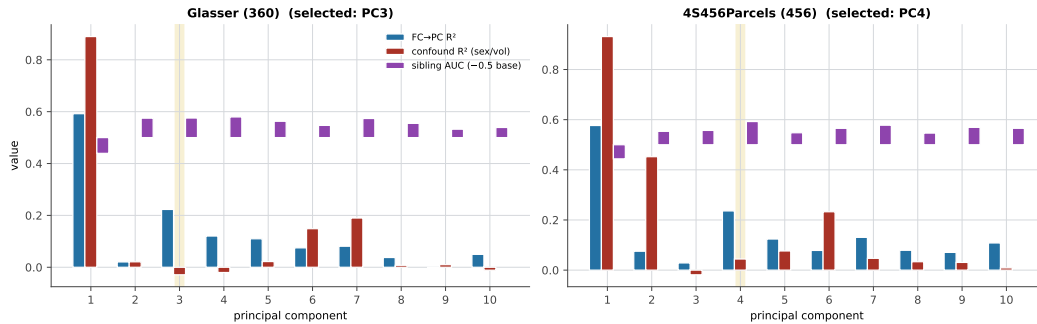


Figure S7: Per-PC FC-predictability, sex/volume confound, and sibling AUC. PC1 is FC-predictable but a confound; the selected low-variance mode is FC-predictable, family-discriminative, and low-confound.

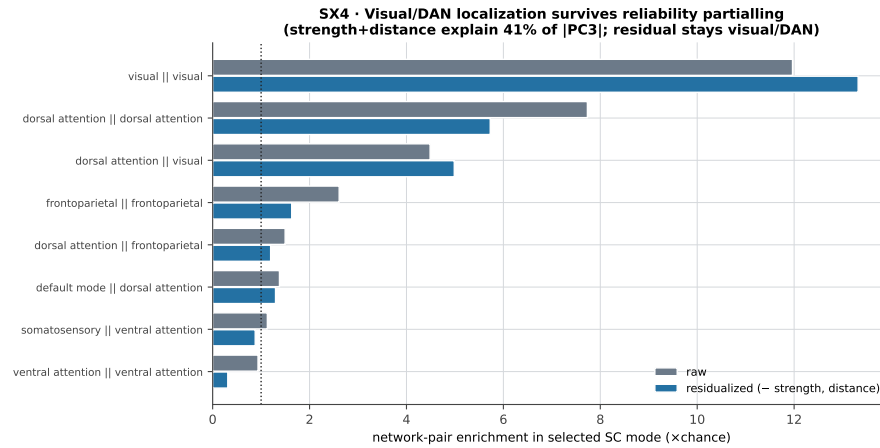


Figure S8: Network-pair enrichment of the selected SC mode, raw vs reliability-residualized.

Source: sanity_checks/tract_check/, reproduction/family_mechanism/outputs/f8_*.csv.

S8 Completeness, Leak Guardrails, and Reproducibility

The spine grid contains **13,640 cells** (2,640 reconstruction + 11,000 downstream); the merged output is complete (13,640/13,640), with zero non-finite values in load-bearing metrics and zero hard leak failures.

Table S12: Leak verdicts separate expected biological signal from implementation leakage.

Verdict	Meaning	Count
ok	no threshold violation	3,326
EXPECTED_SIGNAL	raw connectome predicts sex/age (biological)	4
EXEMPT_FLAGGED	input contains bv+demo; sex/age expected	1,070
LEAK_FAIL	non-exempt derived input exceeds threshold	0

float64 fix to the scalar PCA path. The scalar (downstream) estimators originally ran PCA in float32. The bv+demo input contains exactly-collinear one-hot encodings of sex and race (each dummy set sums to 1), so its feature matrix is rank-deficient; float32 SVD is ill-conditioned on a rank-deficient matrix and scrambled the low-variance directions that carry the demographic signal by a seed-dependent amount (e.g. bv+demo→age fell to 0.46 on one seed instead of ≈1.0). Casting the scalar-path inputs to float64 before PCA (`_grid_common.py`, commit 69add40) recovers those directions exactly. *Blast radius:* only the sex/age leak diagnostics — bv+demo→sex 0.945→1.0, →age 0.840→1.0 (the labels are literally input columns). Cognition (F4/F5) is byte-identical (max $|\Delta\text{lift}| = 0$; baseline 0.359/0.283/0.354), reconstruction is unaffected (bv+demo→SC is float32 ≡ float64 under PLS scaling), F6–F8 do not touch this path, and 0 LEAK_FAIL is preserved.

Estimator-reporting rule. Reconstruction leads with `demeaned_pearson` (raw Pearson is dominated by the population-mean connectome); downstream scalar targets use `BayesianRidge`. A resolved discrepancy showed PCA→PLS scalar rows are ill-conditioned (e.g. Glasser SC→SC oracle `demeaned_pearson` 0.376 under PCA→PLS vs 0.648 under `BayesianRidge`), so every reported number names its estimator. **CSV outputs are the source of truth; W&B is a replay layer, not a dependency.** Family/mechanism ports are validated against the notebooks with exact or near-exact regression tests.

S4 · Leak-check targets (Glasser, BayesianRidge). Sex/age are diagnostics, not scientific outcomes.

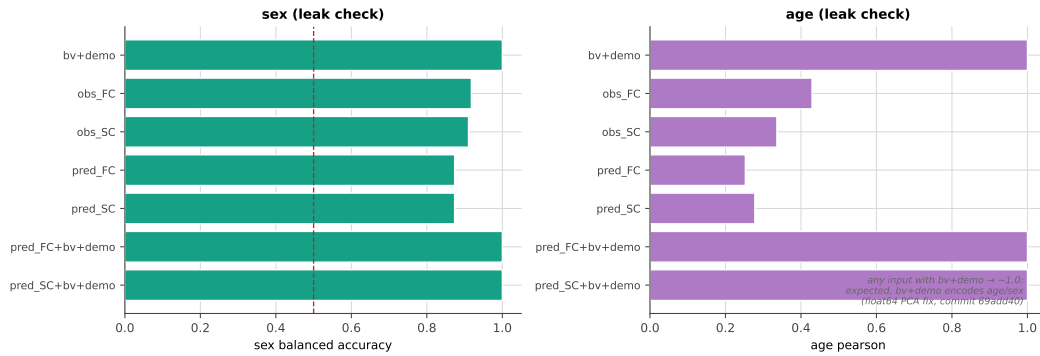


Figure S9: Sex and age prediction by input (Glasser). These are diagnostics, not scientific outcomes; any input containing bv+demo reaches ~1.0 for sex/age because bv+demo literally encodes them (after the float64 PCA fix, these read ≈1.0 consistently across seeds).

Summary — every escape route, closed

Narrowed, defensible claim. In healthy young adults, under cross-sectional normal-range cognition, the missing structural utility is not recovered by obvious changes in dimensionality reduction, model

Table S13: Each mundane explanation for the directional / negative results, and the check that closes it.

Mundane explanation	§	Outcome
The asymmetry is a PCA reduction artifact	S1	ratio > 1 for raw-PLS, PCA, 3 JL; $p \leq 0.001$
SC→FC fails because FC is noisy	S2	SC→FC reaches 17% of FC ceiling; per-subject $r = 0.01$
A bigger / nonlinear model would find it	S3	KernelRidge (validated on FC) finds nothing
The linear model left residual signal unused	S4	OOB residual-boost null; +0.005 recon sliver washes out
More subjects would unlock it	S5	nonlinear gap flat/ ≤ 0 from $n=100$ to $n \approx 683$
Streamline counts are too crude	S6	r2t predicts FC worse, no marginal or cognition gain
The family mode is a reliability artifact	S7	survives strength+distance+reliability partial (41% R^2)
The grid is silently incomplete or leaking	S8	13,640/13,640 cells, 0 non-finite, 0 LEAK_FAIL

class, structural representation, or sample size — and the SC→FC shortfall is not an FC-noise artifact. The remaining open piece is SC’s own reliability, which is data-blocked pending test-retest dMRI.